

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

## Chapter 5: Quadratic Equations

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Check whether  $x = -3$  is a root of  $2x^2 + 5x - 3 = 0$ .
2. If  $x = \frac{2}{3}$  is a root of  $3x^2 + kx - 2 = 0$ , find  $k$ .
3. Solve  $x^2 - 9x + 20 = 0$ .
4. Solve  $6x^2 + x - 12 = 0$ .
5. Solve  $\frac{x+1}{x-1} = \frac{3x-7}{2x-5}$ .
6. Solve  $x^2 - 5x - 10 = 0$ , correct to two decimal places.
7. Solve  $2x^2 - 4x - 3 = 0$ , correct to two decimal places.
8. Find the nature of the roots of  $4x^2 - 12x + 9 = 0$ .
9. Find the value of  $k$  for which  $kx^2 + 6x + 1 = 0$  has equal roots.
10. For which values of  $p$  does  $x^2 + px + 4 = 0$  have real roots?

## Answer Key

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1.  $18 - 15 - 3 = 0$ ; yes
2.  $\frac{4}{3} + \frac{2k}{3} - 2 = 0$ , so  $k = 1$
3.  $x = 4$  or  $x = 5$
4.  $x = \frac{4}{3}$  or  $x = -\frac{3}{2}$
5. It reduces to  $x^2 - 7x + 12 = 0$ , so  $x = 3$  or  $x = 4$
6.  $x = 6.53$  or  $x = -1.53$
7.  $x = 2.58$  or  $x = -0.58$
8.  $D = 0$ : real and equal, each  $\frac{3}{2}$
9.  $36 - 4k = 0$ , so  $k = 9$
10.  $p^2 \geq 16$ :  $p \geq 4$  or  $p \leq -4$